18-MINUTE TEACHER ROUTE ·15–20 MINUTE LESSON

One fixed population.
Many observed estimates.

A compact Predict → Experiment → Observe → Explain → Transfer lesson. Keep the mathematics connected: the population, one complete sample and one statistic from that sample. 参数固定，样本与估计值变化。

Before class: choose one experiment

Open **Sampling & estimators**, then `Teacher mode`. Select **Aurora Archive · skewed** and **Sample mean · $\bar{X}$**. In `3 · One sample`, set **n=3**, seed **aurora-2026**, then `Reset & repeat`. Return to `2 · Fixed population`. Stage changes preserve the sample; model/n/rule changes clear incompatible results.

> **Model:** independent draws with replacement from a fixed weighted card pool. Each sample contains n draws. R counts complete repeated samples. The cards, identities and powers are invented; no payment, owned collection or real student data is involved.

Actual Aurora population · power in points

Power x	2	4	8	16	40
Probability	.45	.30	.16	.07	.02

Preparation checks and seeded values · teacher answer

μ=5.3, σ²=38.67. At n=3, SE($\bar{X}$)=√(38.67/3)≈3.590265. The declared seed starts with [8,40,2], $\bar{x}$=50/3≈16.6667; second $\bar{x}$=16/3≈5.3333. The mean of these two estimates is 11, not 5.3. Reset the same configuration to reproduce; do not reseed between the two draws. If a future implementation changes the stream, use actual values and update this script.

Use Hide worked answers before hand calculation; it conceals calculated output, feedback and graph answers while retaining raw observations and RNG state. Hide true values is separate and changes visibility, not μ. No real-class rehearsal, physical projector-distance check or learning gain is claimed.

The 18-minute route

0–2 MINUTES · PREDICT

01 What stays fixed?

Stage 2 · Fixed population. Show the five possible card types and probabilities.

> “When we draw another group of three, does the pool change? Must every group average the pool mean?”

Have learners name one random object before drawing and one fixed model quantity. Briefly demonstrate replacement: a card type can appear again, even within one group.

Listen for this distinction · teacher answer

The probability model and μ are fixed. X_i is the random power at draw position i before observation. Repeating does not deplete the pool. A possible sample need not have mean μ; a rare 40 is possible and mathematically legitimate.

2–5 MINUTES · EXPERIMENT

02 Three observations become one estimate

Stage 3 · One sample. Check `Hide worked answers`; choose `Draw one sample`. Ask learners to calculate from the three displayed powers, then reveal.

FIXED PARAMETER · 参数	RULE BEFORE SAMPLING · 估计量	OBSERVED RESULT · 估计值
Population mean μ	$\bar{X}=(X_1+X_2+X_3)/3$	**This sample's $\bar{x}$**

> “Is the rule the same object as its numerical answer? Has this sample changed the population mean?”

Expected calculation and language · teacher answer

[8,40,2] gives $\bar{x}=50/3\approx16.667$. $\bar{X}$ is the statistic/estimator before the draw;16.667 is its observed estimate. μ remains 5.3. An unbiased estimator does not promise that the next observed value equals μ.

5–8 MINUTES · OBSERVE

03 One whole group becomes one dot

Choose `Next stage →` to reveal stage 4 · Sampling distribution. Use `Find sample #1's point →`. Draw again; use `Revisit sample` to recall sample1.

> “Does this dot represent one card, or all three cards after applying the rule? Which objects changed on the second draw?”

Explain the two encodings: **bars** show relative frequency in statistic bins; **dots in the separate lower strip** identify complete saved samples. Horizontal dot position is the statistic. Vertical dot separation is layout, not probability.

A useful counterexample · teacher answer

Second $\bar{x} \approx 5.333$. After two draws, the average of the two observed estimates is 11. The theoretical $E(\bar{X}) = 5.3$ is not forced to equal the finite simulation average. At R=1, one occupied histogram bin has100% empirical frequency; its saved dot is still just one sample.

8–11 MINUTES · REPEAT

04 Build the empirical sampling distribution

Select `Repeated samples R +200`, then `Run batch`. Pause if needed; inspect a saved group.

> “What does n count? What does R count? Does adding repetitions make one future group of three less variable?”

Keep the parameter, population, n and estimating rule fixed. Compare **Theoretical E(T)** with **Simulation mean**; their meanings differ even if the numbers happen to be close.

Do not promise finite equality · teacher answer

Starting from R=2, adding 200 yields R=202. n counts observations per sample; R counts independent repetitions of the complete sampling experiment. Increasing R improves Monte Carlo depiction; the theoretical SE of the next estimator is unchanged. Expectation averages over all possible samples with their probabilities.

11–14 MINUTES · EXPLAIN

05 Exact possibilities have unequal weights

Stay at stage 4. Change to n=2; explain that the old cohort clears. Enable `Exact small model`, then expand `Inspect exact ordered samples & weights`.

> “There are 25 ordered pairs. Are they equally likely? Compare the two-lowest and two-rarest pairs.”

Inspect [2,2] on **Page 1**. Set the inspector’s `Page` field to **3** for [40,40], case 25. Compare the total beneath the table, the original values, product probability and weighted contribution. If the full table overloads the group, compare just two rows and the exact total.

Weighted exact expectation · teacher answer

[2,2] has probability.45^2=.2025, mean=2 and contribution.405. [40,40] has probability.02^2=.0004, mean=40 and contribution.016. All25 probabilities sum1; the weighted sum of means is 5.3. Averaging unique mean values equally would describe a different model. Exact enumeration is capped; beyond the cap the UI does not claim an exact result. S^2 at n=1 is undefined.

14–16 MINUTES · TRANSFER

06 Separate n from R

Return to n=3 and record the theoretical SE before changing n. Then compare fresh n=12. If comparing empirical plots, use the same stated R for both fresh cohorts. The core two-minute task is the theoretical-SE comparison; extra simulation is optional. Ask for a prediction before changing n rather than judging width from a few random points.

> “What happens to μ, E($\bar{X}$) and SE when we quadruple n? What if we only quadruple R?”

The theoretical comparison · teacher answer

μ and E($\bar{X}$) stay 5.3. SE=σ/√n halves:3.590265→1.795132. Changing R alone leaves theoretical SE unchanged. The original skewed population stays skewed. Its sample mean is not exactly Normal for every small n, and n=30 is not a universal guarantee.

16–18 MINUTES · EXIT CHECK

07 Ask for an explanation, not a button sequence

Use questions1–3 from the student transfer sheet below. Ask learners to identify what exists before drawing, what changes next time, and what one point means. A correct choice with no reason is incomplete evidence. If time allows, use question=4 to compare a wasteful but unbiased rule with the mean.

CLT and repeated confidence coverage are follow-on lessons. Do not add their full explanations to this 18-minute demonstration.

ORIGINAL TRANSFER · STUDENT SHEET

Can you use the idea in a new model?

New fictional model: each independent draw returns1 point with probability.75 or5 points with probability.25. Sampling is with replacement. We use n=4 draws to estimate the fixed population mean μ. A realised sample is **[1,1,5,1]**. These questions are original teaching material, not copied exam questions.

1. Find μ. Identify a parameter, an estimator before drawing and its observed estimate for this realised sample. What does the next dot in the mean-distribution strip represent?

2. We repeat this four-draw experiment 1000 rather than=100 times. Which changes: the theoretical SE of one future sample mean, the Monte Carlo depiction, both or neither? Give a reason.

3. Keep the rule $\bar{X}$, but change n=4 to n=16. Describe the population mean, $E(\bar{X})$ and SE. Does the next estimate have to equal μ?

4. Compare rules $T_1=\bar{X}$, $T_2=X_1$ and $T_3=\bar{X}+1$ as estimators of μ. Which are unbiased? Does using all four observations automatically remove the offset in T_3? Optional: calculate each theoretical SE.

5. For n=2, list the four ordered samples. Give their probabilities and show that the probability-weighted expected sample mean equals μ. Explain why four equally weighted cases would be wrong.

6. Follow-on CI question: after an interval is computed, does “95% confidence” assign 95% probability to fixed μ lying in that realised interval? Must exactly 95 out of 100 repeated intervals cover? Explain both claims.

Teacher answers · original transfer sheet · teacher answer

1. μ=.75×1+.25×5=2. Parameterμ; estimator$\bar{X}$; observed estimate(1+1+5+1)/4=2. One dot is the mean computed from one complete sample of four observations, not an individual observation. Equality here is coincidence, not a requirement.
2. Only the Monte Carlo depiction becomes more precise. n/model/rule determine the distribution of one estimator; R counts how often we simulate it.
3. μ and E($\bar{X}$) stay 2. SE halves when n quadruples. The next observed estimate need not equal2.
4. $E(T_1)=E(T_2)=2$, so both unbiased. $E(T_3)=3$, bias+1; larger n does not remove it. Population variance=$.75(1-2)^2+.25(5-2)^2=3$. At n=4: $SE(T_1)=SE(T_3)=\sqrt{3}/2\approx.866025$; $SE(T_2)=\sqrt{3}\approx1.732051$. Adding a constant changes expectation, not variance.
5. [1,1],[1,5],[5,1],[5,5] have probabilities9/16,3/16,3/16,1/16 and means1,3,3,5. Weighted expectation=(9+9+9+5)/16=2. Equal weighting would give 3 and ignore the declared probabilities.
6. No to both claims. In this frequentist model μ is fixed; repeated samples give random endpoints. The method's nominal coverage concerns repetitions under its conditions; the realised interval either covers or misses. Finite coverage fluctuates, and approximation error can matter. Unbiasedness alone does not guarantee95% interval coverage.

Sources, scope and honest limits

Core route: Pearson Edexcel IAL S2.3 §4.1–4.2 and S3.3 §3.1–3.2. Only the official specification defines assessed scope. Card art, exact enumeration, repeated-coverage simulation and design-SE support are pedagogical representations; some explanatory features are enrichment.

Official specification · Issue3 April2019 · S1 printed 55–56, S2 58–59, S3 61–62 (one-based PDF page = printed page+6).

Official Formula Book · Issue2 January2021 · correlation tables printed 27 / PDF 33. See current coverage and method conditions for partial and absent areas.

This script follows the actual controls and independently checked Aurora seed. It is not evidence of a real classroom trial, measured18-minute completion, student learning gain, screen-reader speech or physical projection readability. All data are synthetic. The runtime computes locally and uses no account, payment or student-data upload.
